EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)

Topic	**D: Further Algebra and Functions**
Sub topic	**D9: Rational Inequalities**

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic	**D: Further Algebra and Functions**
Sub topic	**D9: Rational Inequalities**

To be used by all students and teachers for AQA A Level Further Mathematics (7367) for exams from 2025 onwards.

Students and teachers of other boards may also find this useful

Name : ____________________

Date : ____________________

Time : ____________________

Total Marks Available : _____

Total Marks Received : _____

Q1.

The curve C has equation $y = \frac{x}{(x+1)(x-2)}$.

The line L has equation $y = -\frac{1}{2}$.

(a) Write down the equations of the asymptotes of C.

(3)

(b) The line L intersects the curve C at two points. Find the x-coordinates of these two points.

(2)

(c) Sketch C and L on the same axes.

(You are given that the curve C has no stationary points.)

(3)

(d) Solve the inequality

$$\frac{x}{(x+1)(x-2)} \leq -\frac{1}{2}$$

(3)

(Total 11 marks)

Q2.

A curve has equation $y = \frac{1}{x^2 - 4}$.

(a) (i) Write down the equations of the three asymptotes of the curve.

(3)

(ii) Sketch the curve, showing the coordinates of any points of intersection with the coordinate axes.

(4)

(b) Hence, or otherwise, solve the inequality

$$\frac{1}{x^2 - 4} < -2$$

(3)

(Total 10 marks)

Q3.

(a) (i) Write down the equations of the two asymptotes of the curve $y = \frac{1}{x-3}$.

(2)

(ii) Sketch the curve $y = \dfrac{1}{x-3}$, showing the coordinates of any points of intersection with the coordinate axes.

(2)

(iii) On the same axes, again showing the coordinates of any points of intersection with the coordinate axes, sketch the line $y = 2x - 5$.

(1)

(b) (i) Solve the equation

$$\frac{1}{x-3} = 2x - 5$$

(3)

(ii) Find the solution of the inequality

$$\frac{1}{x-3} < 2x - 5$$

(2)

(Total 10 marks)

Q4.

A curve has equation

$$y = \frac{(x-1)(x-3)}{x(x-2)}$$

(a) (i) Write down the equations of the three asymptotes of this curve.

(3)

(ii) State the coordinates of the points at which the curve intersects the x-axis.

(1)

(iii) Sketch the curve.

(You are given that the curve has no stationary points.)

(4)

(b) Hence, or otherwise, solve the inequality

$$\frac{(x-1)(x-3)}{x(x-2)} < 0$$

(2)

(Total 10 marks)

Q5.

A curve has equation

$$y = \frac{x}{x^2 - 1}$$

(a) Write down the equations of the three asymptotes to the curve.

(3)

(b) Sketch the curve.

(You are given that the curve has no stationary points.)

(4)

(c) Solve the inequality

$$\frac{x}{x^2 - 1} > 0$$

(3)

(Total 10 marks)

Q6.

A curve has equation

$$y = \frac{3x - 1}{x + 2}$$

(a) Write down the equations of the two asymptotes to the curve.

(2)

(b) Sketch the curve, indicating the coordinates of the points where the curve intersects the coordinate axes.

(5)

(c) Hence, or otherwise, solve the inequality

$$0 < \frac{3x - 1}{x + 2} < 3$$

(2)

(Total 9 marks)

Q7.

A curve has equation

$$y = \frac{6x}{x - 1}$$

(a) Write down the equations of the two asymptotes to the curve.

(2)

(b) Sketch the curve and the two asymptotes.

(4)

(c) Solve the inequality

$$\frac{6x}{x-1} < 3$$

(4)
(Total 10 marks)